MANAGERIAL ECONOMICS

Course Overview

This course introduces M.Tech students to the principles of managerial economics and their application in technology-driven business environments. It equips students with analytical tools to understand market structures, firm behaviour, pricing strategies, demand-supply dynamics, and decision-making under uncertainty.

The course bridges engineering and business thinking, enabling students to evaluate the economic consequences of technical decisions, understand the competitive landscape of technology markets, and develop a foundation for entrepreneurship, product management, and technology leadership roles.

Course Structure & Syllabus

1. Foundations of Managerial Economics
   - Nature and scope of managerial economics; its relation to microeconomics and decision science
   - Scarcity, choice, and opportunity cost in technology resource allocation
   - Marginal analysis and optimisation: the core decision-making framework
   - Economic profit vs. accounting profit; the role of economic reasoning in engineering decisions
   - Time value of money: present value and future value basics
   - Case: Resource allocation decisions in a software product team

2. Demand Analysis & Forecasting
   - Law of demand; demand functions and demand curves for technology products
   - Determinants of demand: price, income, substitutes, complements, and network effects
   - Price elasticity of demand: concept, measurement, and managerial applications
   - Income elasticity and cross-price elasticity
   - Demand for digital goods: zero marginal cost, bundling, and versioning
   - Demand forecasting methods: qualitative (Delphi, expert opinion) and quantitative (time series, regression)
   - Case: Estimating demand elasticity
   - Demand estimation using real datasets (industry/API data)
   - Machine learning approaches to forecasting (basic intro)

3. Production & Cost Analysis
   - Production functions: short run vs. long run; total, average, and marginal product
   - Law of diminishing marginal returns and its implications for scaling tech teams
   - Isoquants and isocosts; optimal input combinations
   - Cost concepts: fixed, variable, total, average, and marginal costs
   - Economies of scale, economies of scope, and learning curve effects in technology industries
   - Cost structure of software and AI firms: high fixed costs, near-zero marginal costs
   - Break-even analysis and contribution margin

- Case: Cost analysis of cloud vs. on-premise infrastructure deployment
- Productivity analytics (TFP, efficiency frontier)
- Cost optimization using operations research techniques
- Capacity planning in cloud/AI systems

4. Market Structures & Competitive Strategy
   - Overview of market structures: perfect competition, monopolistic competition, oligopoly, monopoly
   - Profit maximisation under different market structures
   - Monopoly and market power: sources, welfare implications, and regulation
   - Monopolistic competition and product differentiation in technology markets
   - Oligopoly models: Cournot, Bertrand, and Stackelberg competition
   - Digital markets and platform economics: two-sided markets, network effects, and winner-take-all dynamics
   - Antitrust and competition policy in the technology sector
   - Porter's Five Forces applied to technology industries
   - Case: Market structure analysis of cloud computing, ride-hailing, and search engine markets

5. Pricing Strategies in Technology Markets
   - Pricing objectives and the role of elasticity in pricing decisions
   - Cost-plus pricing, value-based pricing, and competitive pricing
   - Price discrimination: first, second, and third degree; implementation in digital platforms
   - Freemium models, tiered pricing, and subscription economics
   - Dynamic pricing and algorithmic pricing: applications in e-commerce and ride-sharing
   - Bundling strategies: pure bundling, mixed bundling, and tying
   - Penetration pricing vs. skimming pricing for new technology products
   - Transfer pricing and pricing in multi-division technology firms
   - Case: Pricing strategy design for an AI-as-a-service product

6. Game Theory & Strategic Decision-Making
   - Introduction to game theory: players, strategies, payoffs, and Nash equilibrium
   - Prisoner's dilemma and its applications to technology competition (price wars, patent races)
   - Dominant strategies and iterated elimination of dominated strategies
   - Sequential games and backward induction; first-mover and second-mover advantages
   - Repeated games and cooperation: implications for strategic alliances and open-source ecosystems
   - Mixed strategies and equilibrium
   - Bayesian games (incomplete information)
   - Mechanism design basics (auctions, incentives)
   - Applications in: ad auctions (Google Ads), spectrum auctions, and platform competition

7. Macroeconomics & the Business Environment
   - Key macroeconomic indicators: GDP, inflation, unemployment, and their relevance to technology business planning
   - Business cycles and their effect on technology investment and hiring
   - Monetary policy and interest rates: impact on technology firm valuations and capital costs
   - Fiscal policy, government spending on R&D, and industrial policy for technology sectors

- Foreign exchange rates and globalisation of technology supply chains
- India's digital economy: key indicators, policy frameworks (Digital India, PLI schemes), and growth drivers
- Case: Macro-environment analysis for expanding a technology startup internationally

Core Textbooks

1. Samuelson, W. F., & Marks, S. G. (2021). Managerial Economics (8th ed.). Wiley.

2. Pindyck, R. S., & Rubinfeld, D. L. (2018). Microeconomics (9th ed.). Pearson.

3. Salvatore, D. (2020). Managerial Economics in a Global Economy (9th ed.). Oxford University Press.

4. Keat, P. G., Young, P. K. Y., & Erfle, S. E. (2013). Managerial Economics: Economic Tools for Today's Decision Makers (7th ed.). Pearson.

5. Froeb, L. M., McCann, B. T., Ward, M. R., & Shor, M. (2018). Managerial Economics: A Problem-Solving Approach (5th ed.). Cengage Learning.

Evaluation Scheme

25% Assignment – First Assignment
25% Mid – term Assignment
50% - End sem (Term Paper)